

A.2 Section 2

Remark A.1 (An advanced mathematical remark^{*1}). Let $\mathcal{F}$ be the set of all events. For example, we always have $\emptyset, \Omega \in \mathcal{F}$. Then $\mathcal{F}$ is a subset of the *power set* 2^Ω of Ω (by definition 2^Ω is the set of all subsets of Ω). In fact, in many situations one should not, and also cannot, take $\mathcal{F}$ to be 2^Ω . In other words, some subsets of Ω may not be events. An aspect of this idea is illustrated with some examples in Section A.2.1. To be fully rigorous, the collection $\mathcal{F}$ should be given separately (as the domain of the probability measure $\mathbb{P}$). Also, $\mathcal{F}$ is required to satisfy the following properties: (i) $\Omega \in \mathcal{F}$; (ii) if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$; (iii) if $(A_i)_{i \geq 1}$ is a sequences of sets in $\mathcal{F}$, then $\bigcup_{i \geq 1} A_i \in \mathcal{F}$. In words, $\mathcal{F}$ contains Ω and is *closed* under complements and *countable* unions. Such a collection is called a σ -*algebra* on Ω . To summarize, to specify an experiment we should specify a pair $(\Omega, \mathcal{F})$ where Ω is the sample space and $\mathcal{F}$ is a σ -algebra on Ω . Sets in $\mathcal{F}$, and only those, are called *events*. Also, strictly speaking, a probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a sample space, $\mathcal{F}$ is the collection of events (which forms a σ -algebra), and $\mathbb{P}$ is a probability measure regarded as a function on $\mathcal{F}$: $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$. In our course, when we specify a sample space it is understood that the σ -algebra has been implicitly specified in such a way that the sets considered are events.

A.2.1 Exercises

1. Draw a graph to illustrate the event A in Example 2.3. Also illustrate graphically the event $B = \{\omega \in \Omega : |\omega_1 - \omega_2| \geq 0.1\}$, i.e., the two measurements are at least 0.1 apart.
2. Suppose an urn contains three balls: one red (R), one green (G), and one blue (B). Apart from the colours they are identical. A ball is to be picked from the urn. Let the sample space be $\Omega = \{R, G, B\}$. However, the observer is colour-blind and cannot distinguish the red ball from the green ball. That is, the observer can only tell us whether the ball is (i) “blue” or (ii) “red or green”. Write down all events whose occurrence (or not) can be determined by the observer. (There are 4 such events.)
3. Suppose a coin is tossed twice. Let the sample space be $\Omega = \{++, +-, -+, --\}$. Write down *all* events whose occurrence (or not) can be determined by observing *only* the total number of +. (There are 8 such events.)
4. Is it possible to define a “uniform distribution” on $\mathbb{N} = \{1, 2, \dots\}$, the set of natural numbers? Explain.
5. Show that if $\mathbb{P}(A \triangle B) = 0$ then $\mathbb{P}(A) = \mathbb{P}(B)$. (Recall that $A \triangle B$ is the symmetric difference.)
6. Show that A and B are independent if and only if A^c and B are independent. Note that this implies that

$$A \perp\!\!\!\perp B \Leftrightarrow A^c \perp\!\!\!\perp B \Leftrightarrow A \perp\!\!\!\perp B^c \Leftrightarrow A^c \perp\!\!\!\perp B^c. \quad (\text{A.1})$$

¹Sections/remarks marked with an asterisk deal with more advanced concepts and may be skipped on first reading. The materials treated here will *not* be tested in this course.

7. Using the information given in Example 2.30, compute by brute force the probability mass function of the random variable X there. That is, implement directly the formula

$$\mathbb{P}(X = x) = \sum_{\omega \in \Omega: X(\omega) = x} \mathbb{P}\{\omega\},$$

for $x = 0, 1, \dots, 5$. In words, for each x , enumerate all outcomes ω such that $X(\omega) = x$, and sum the corresponding probabilities. The computation is tedious if done by hand, so you should use a computer. The proper approach, which involves the *binomial coefficient*, will be given later. The following R script may be used to start:

```
## construct the sample space
number2binary = function(number, noBits) {
  # a function to produce binary numbers (use as a blackbox)
  binary_vector = rev(as.numeric(intToBits(number)))
  if(missing(noBits)) {
    return(binary_vector)
  } else {
    binary_vector[-(1:(length(binary_vector) - noBits))]
  }
}

n <- 5
Omega <- character(2^n)
for (i in 1:2^n) {
  Omega[i] <- paste(number2binary(i - 1, n), collapse = "")
}
Omega # print and see

## compute X(omega) for all omega
library(stringr)
X <- str_count(Omega, "1")

## Begin by computing P(omega) for all omega
# ...
```